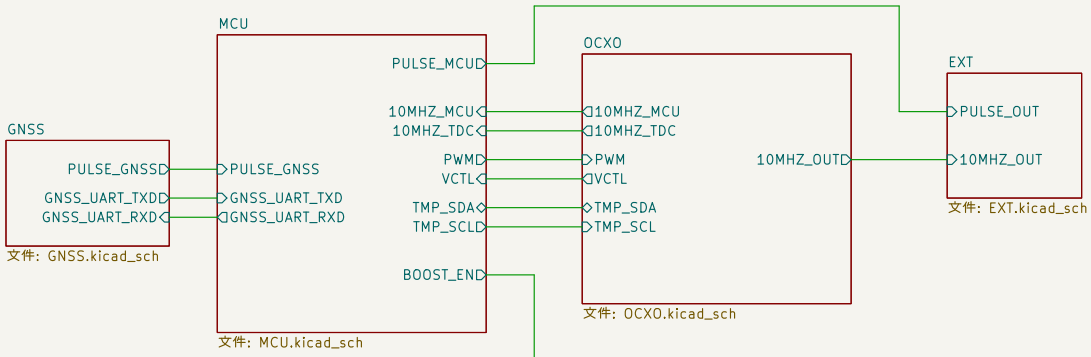
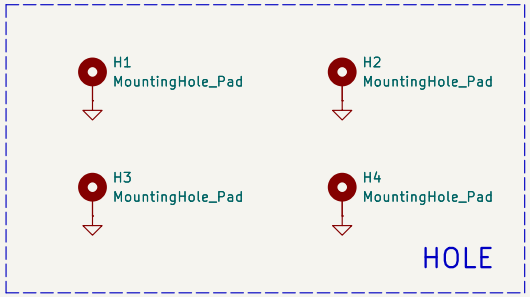


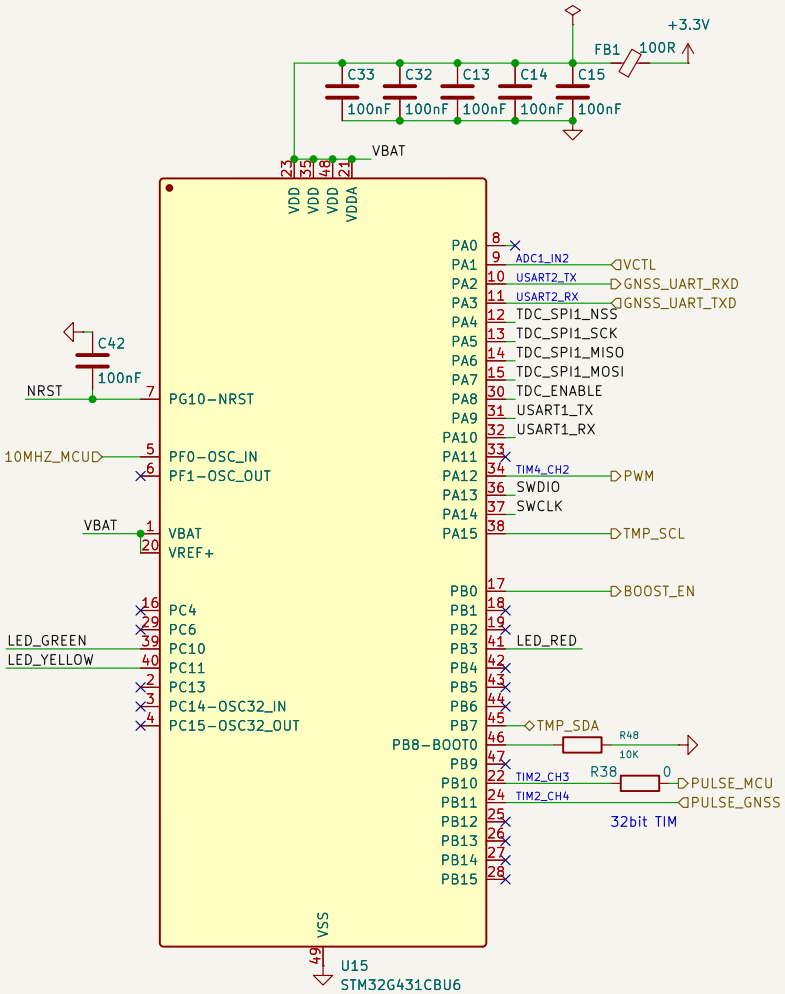


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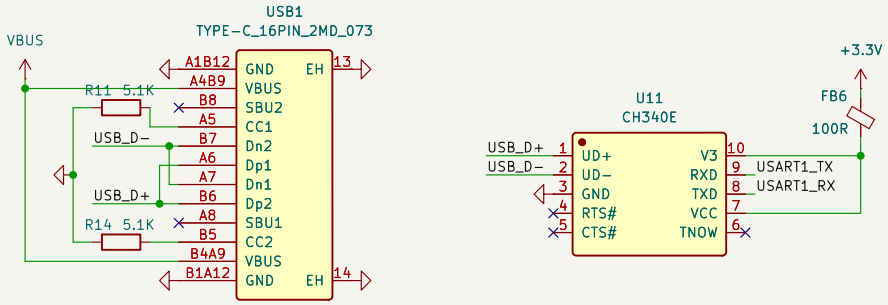
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/6

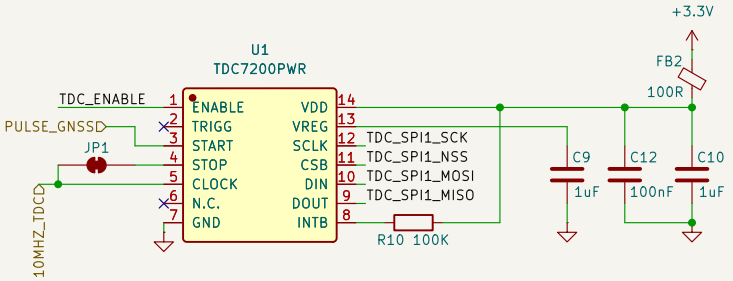
Microcontroller



USB Connector



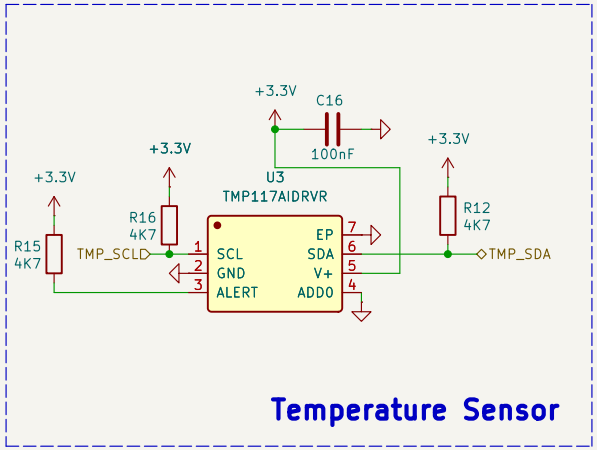
Time-to-Digital Converter

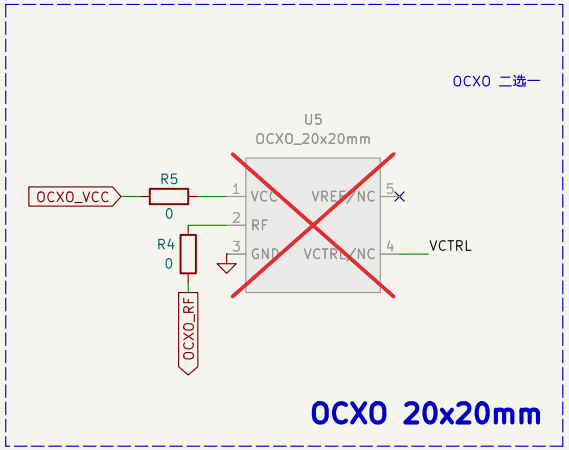


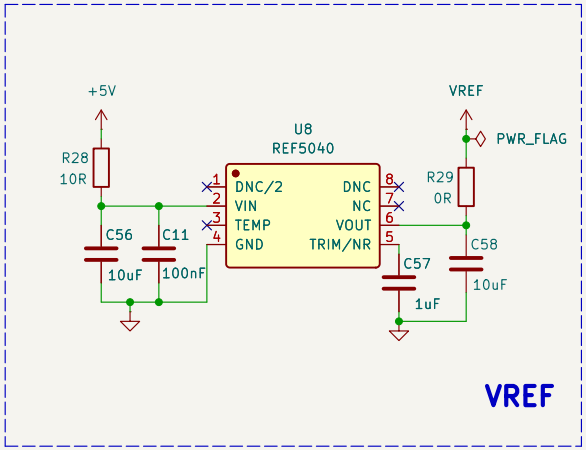
SWD Connector

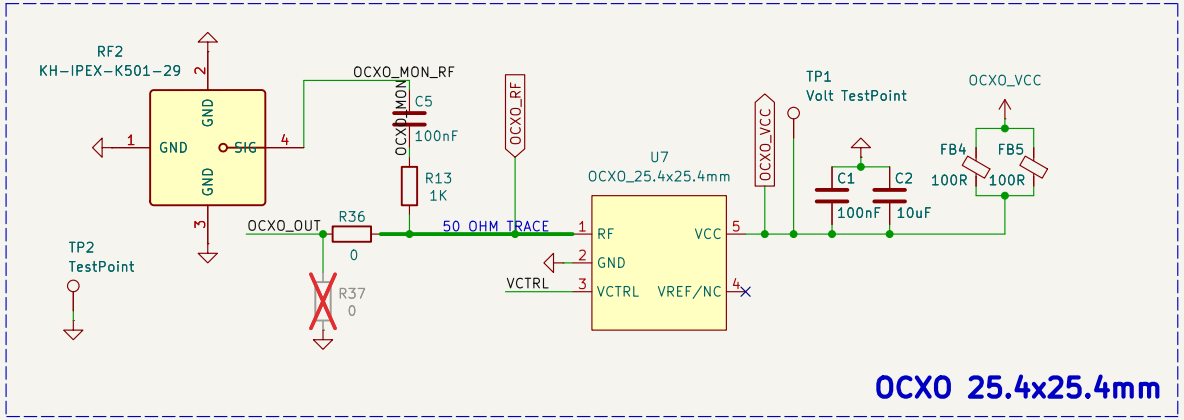


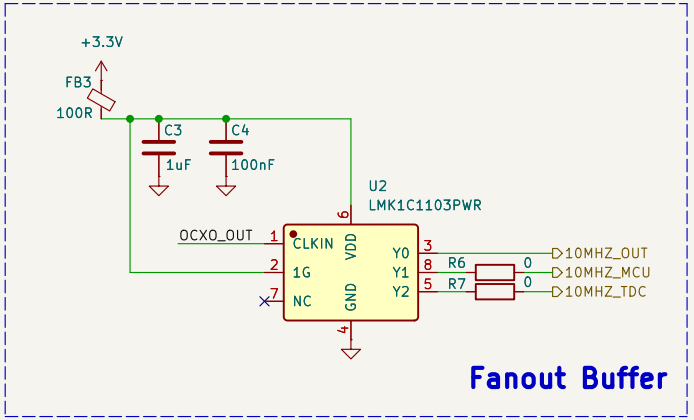
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/6



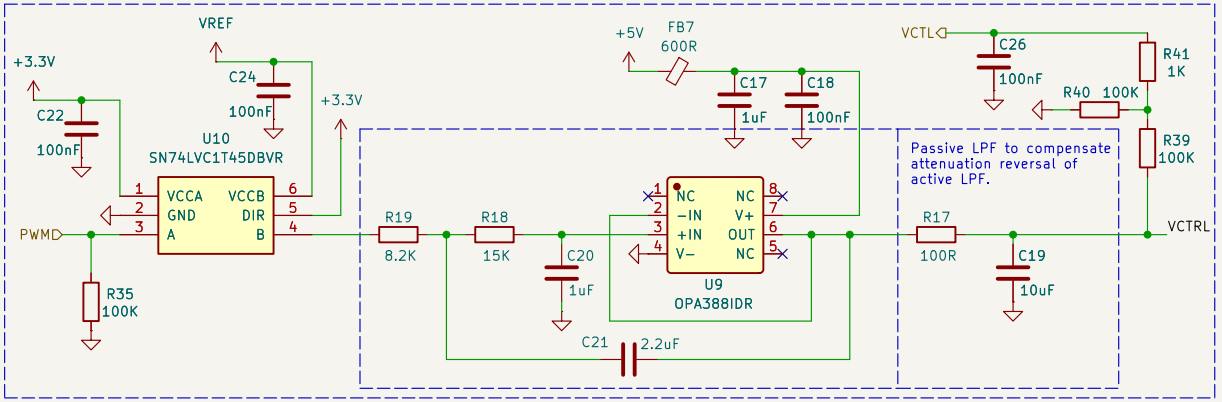








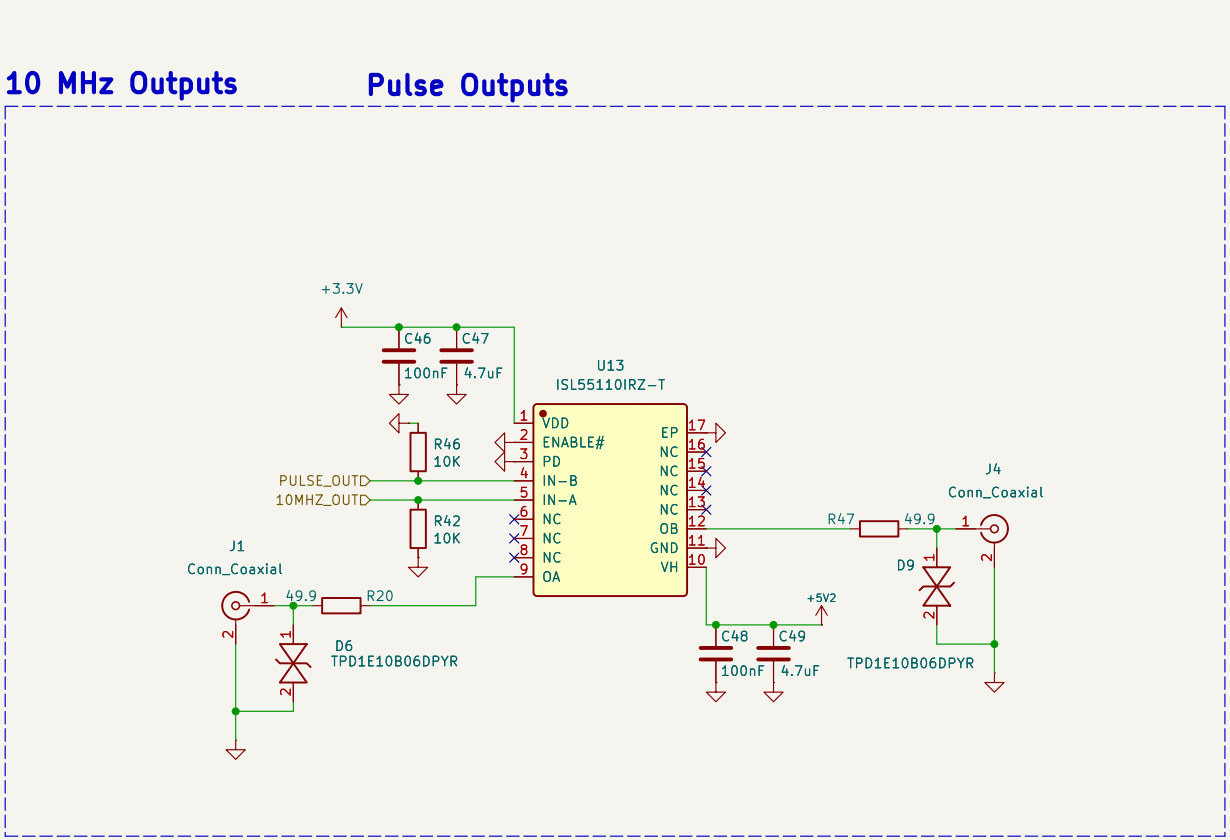
Active Low-pass Filter for Tuning Voltage



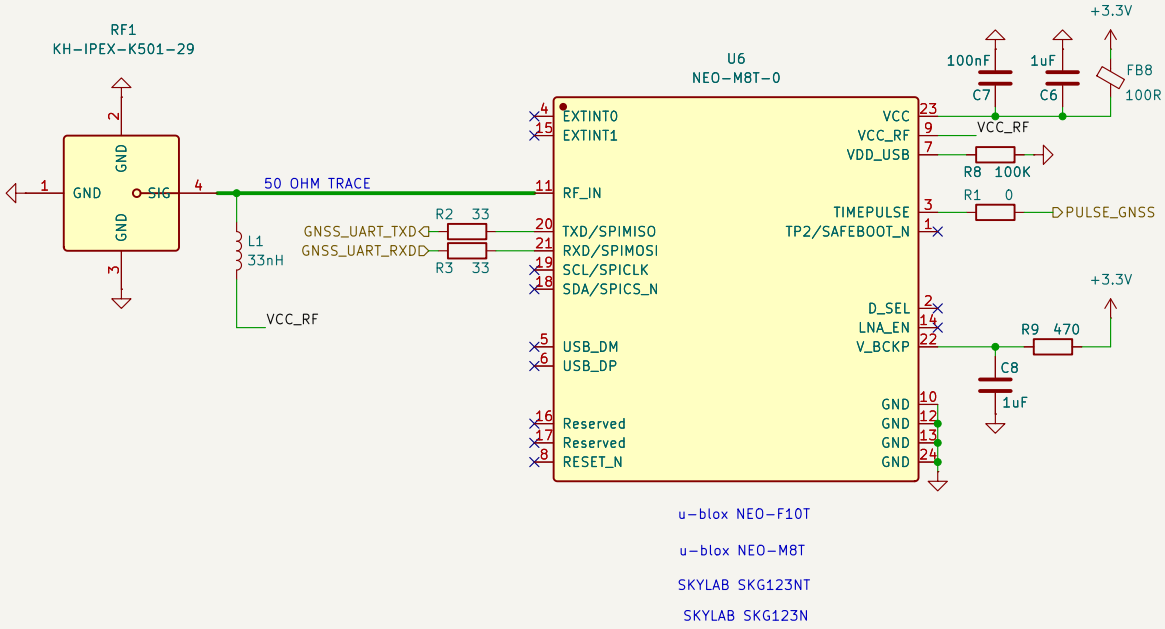
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/6

10 MHz Outputs

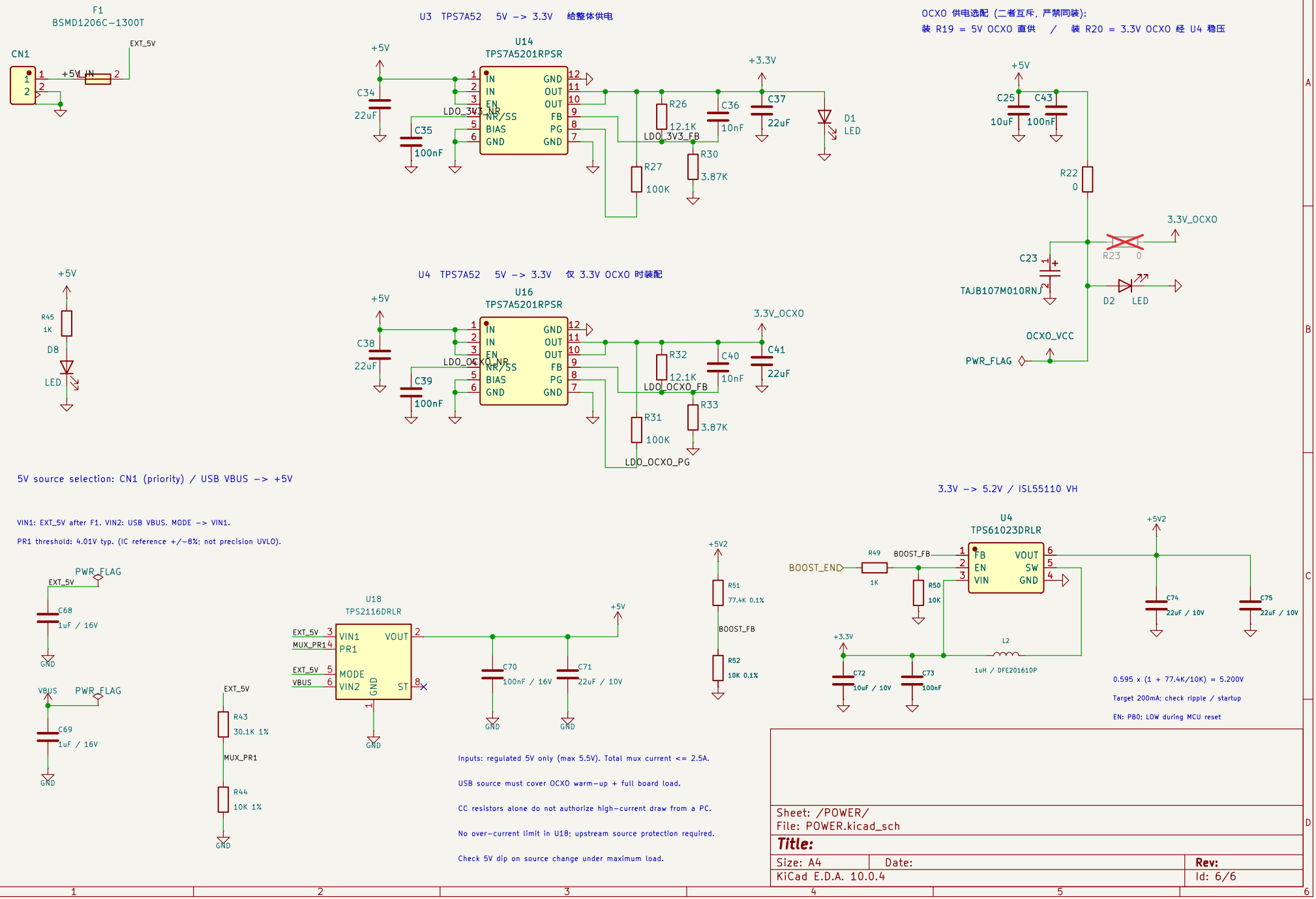
Pulse Outputs



Sheet: /EXT/		
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/6



Sheet: /GNSS/		
File: GNSS.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/6


F1
BSMD1206C-1300T
CN1
EXT_5V
+5V_IN
U3 TPS7A52 5V -> 3.3V 给整体供电
U14
TPS7A5201RPSR
+5V
+3.3V
C34
22uF
C35
100nF
LDO_3V3_NR
R26
12.1K
LDO_3V3_FB
C36
10nF
C37
22uF
D1
LED
R27
100K
R30
3.87K
OCXO 供电选配 (二者互斥, 严禁同装):
装 R19 = 5V OCXO 直供 / 装 R20 = 3.3V OCXO 经 U4 稳压
C25
10uF
C43
100nF
R22
0
3.3V_OCXO
R23
0
C23
TAJB107M010RNJ
D2
LED
OCXO_VCC
PWR_FLAG
R45
1K
D8
LED
U4 TPS7A52 5V -> 3.3V 仅 3.3V OCXO 时装配
U16
TPS7A5201RPSR
C38
22uF
C39
100nF
LDO_OCXO_NR
R32
12.1K
LDO_OCXO_FB
C40
10nF
C41
22uF
R31
100K
R33
3.87K
LDO_OCXO_PG
5V source selection: CN1 (priority) / USB VBUS -> +5V
VIN1: EXT_5V after F1. VIN2: USB VBUS. MODE -> VIN1.
PR1 threshold: 4.01V typ. (IC reference +/-8%; not precision UVLO).
PWR_FLAG
EXT_5V
C68
1uF / 16V
GND
VBUS
PWR_FLAG
C69
1uF / 16V
GND
U18
TPS2116DRLR
EXT_5V
MUX_PR1
VBUS
VIN1
VIN2
PR1
MODE
VOUT
GND
ST
C70
100nF / 16V
C71
22uF / 10V
+5V
EXT_5V
R43
30.1K 1%
MUX_PR1
R44
10K 1%
GND
+5V2
R51
77.4K 0.1%
BOOST_FB
R52
10K 0.1%
3.3V -> 5.2V / ISL55110 VH
U4
TPS61023DRLR
BOOST_EN
R49
1K
BOOST_FB
R50
10K
FB
EN
VIN
VOUT
SW
GND
+3.3V
L2
1uH / DFE201610P
C72
10uF / 10V
C73
100nF
+5V2
C74
22uF / 10V
C75
22uF / 10V
0.595 x (1 + 77.4K/10K) = 5.200V
Target 200mA; check ripple / startup
EN: PB0; LOW during MCU reset
Inputs: regulated 5V only (max 5.5V). Total mux current <= 2.5A.
USB source must cover OCXO warm-up + full board load.
CC resistors alone do not authorize high-current draw from a PC.
No over-current limit in U18; upstream source protection required.
Check 5V dip on source change under maximum load.
Sheet: /POWER/
File: POWER.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 6/6